Nathan Zhang

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EDUCATION

The University of British Columbia Sep 2023 – May 2028
B.A., Combined Major in Economics and Statistics, Minor in Data Science **Average: 93.5/100**

- **Coursework:** Statistical Inference, Machine Learning and Data Mining, Causal Methods, Advanced Econometrics, Statistical Learning, Time Series and Forecasting, Computational Optimization
- **Awards:** Work Learn International Undergraduate Research Awards (2026), Economics Undergraduate Scholarship (2025), Faculty of Arts International Student Scholarship (2024, 2025)

EXPERIENCE

ESG Research Analyst Co-op Sep 2026 – Present
British Columbia Investment Management Corporation (BCI), Capital Markets & Credit Investments

- Apply causal machine learning to investigate how executive compensation and other governance risk factors affect financial performance, supporting risk-factor screening for fundamental investment research.
- Independently build reproducible Databricks ETL pipelines from Capital IQ/ISS ECA data, construct modeling variables, and develop ML models to predict M&A and other corporate events.
- Backtest voting strategies against historical votes with simulated ballot-time information, and analyze correlations with subsequent risk indicators to inform proxy voting.

Research Assistant Apr 2026 – Present
UBC Sauder School of Business

- Study technological innovation with PatentsBERTa patent/task embeddings, validated against Qwen and manual reading. As sole RA for J. Frank Li and Cheng Chen, fit weighted FE OLS/GMM wage and sorting models, run large-scale estimation and bootstrap inference on Bash/Slurm HPC, and draft the empirical section.
- Prepared financial alternative data from 3.2M+ ProQuest articles and video/audio metadata. Independently built reproducible ETL for RavenPack coverage checks, metadata-to-panel parsing, media archiving, deduplication, and subtitle validation (Allen Hu, May-Aug 2026).

Research Assistant Jan 2025 – May 2026
Vancouver School of Economics at UBC

- Developed data science and ML teaching/research projects in computer vision, NLI, and sentiment analysis. Led proposals, implemented image-embedding/classification workflows, co-developed human-validated zero-shot stance analysis, and designed and taught a CV workshop.

PUBLICATIONS & MANUSCRIPTS

Sequential Tone Residualization and Abnormal Returns in Earnings Calls 2026
Joint first author, accepted at ICAIF 2026 and selected for oral presentation

- Proposed the earnings-call study and co-designed sequential information sets for tone residualization using financial controls and prior dialogue. Built the Capital IQ/IBES/CRSP panel and proposed dictionary/LLM validation.
- On 1,901 held-out 2025 calls, answer-tone residuals were positively associated with market-adjusted returns, with pre-event falsification checks.

Memory Response Paths and Path-Structured Context Compilation 2026
Joint first author, submitted to AAAI and under review

- Proposed Memory Response Paths to trace how memories affect LLM-agent decisions. Designed matched availability experiments and context-rendering ablations, built the 180-ecology benchmark and model API workflows, and conducted statistical analysis.
- A separate 144-item evaluation found PSCC improved LLM-judged quality by 6.9 percentage points over A-Mem with roughly one-third fewer API tokens.

TECHNICAL SKILLS

Programming & ETL: Python, R, SQL, PySpark, DuckDB, Polars, Databricks, Bash, HPC/Slurm
Quantitative methods: panel econometrics, GMM, causal inference, time-series forecasting, machine learning
Text, image & multimodal: NLP, semantic embeddings, computer vision, multimodal ETL
Modeling & finance tools: PyTorch, scikit-learn, fixest, Bloomberg, BQL, Git